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Exploitative and exploratory innovations in knowledge network and collaboration network: A patent analysis in the technological field of nano-energy

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ABSTRACT

Innovations of organizations are doubly embedded in knowledge networks constituted by coupling among knowledge elements and in social networks formed by collaborative relationships among organizations. This study explores the structural properties of such relationships and their possible influences on organizational innovations in terms of exploitation and exploration in the emerging nano-energy field. Results indicate that the knowledge networks and the technology-based collaboration networks in the nano-energy field are decoupled and that they have different degrees of integration. Some structural features of knowledge and collaboration networks influence organizations' exploitative and exploratory innovations in diverse ways. Firstly, direct ties of an organization's knowledge elements in a knowledge network have an inverted U-shaped effect on its exploitative innovation, which is not the case in exploratory innovation. Direct ties in a collaboration network have an inverted U-shaped effect on both its exploitative and exploratory innovations. Secondly, indirect ties of an organization's knowledge elements in a knowledge network affect its exploitative innovation, but not its exploratory innovation. However, indirect ties in a collaboration network affect exploratory innovation, but not exploitative innovation. Thirdly, non-redundancy among ties in a knowledge network exhibits the opposite effect, hindering exploitative innovation, but favoring exploratory innovation. By contrast, non-redundancy among ties in a collaboration network favors exploitative innovation, but shows a non-significant effect on exploratory innovation.

1. Introduction

Innovation is a collective and social activity. A considerable and fast-growing body of empirical research has shown that innovation by individuals or higher-level collectives (i.e., teams, organizations or countries) is influenced by their social relationships and the networks they constitute by enabling or constraining them to acquire, transfer, absorb, evaluate and apply knowledge and information (Demirkan and Demirkan, 2012; Gonzalez-Brambila et al., 2013; Guan and Zhao, 2013; Guan et al., 2015; Vanhaverbeke et al., 2006). However, in reality, an innovation by individuals or higher-level collectives is embedded not only in social networks but also in knowledge networks (Wang et al., 2014; Yayavaram and Ahuja, 2008). Knowledge elements or components form associational relationships with one another in innovative processes lead to the formation of a knowledge network in which their past combinatorial relationships are recorded (Phelps et al., 2012; Yayavaram and Ahuja, 2008). Investigation on the critical roles played by both social and knowledge networks needs to be attained a full understanding of the antecedents of innovation performance (Wang et al., 2014). However, to date, few studies have examined the effects of knowledge networks on innovation outcomes of individuals or higher-level collectives, let alone the integration of social and knowledge networks into an analysis framework. Our study is designed to gain further understanding about why and how the relational and structural properties of both social and knowledge networks facilitate and constrain exploitative and explorative innovations of organizations.

Social networks such as technology-based collaboration network or science-based co-authorship reflect the social interactions among a set of agents—individuals, teams, organizations or even countries (Cantner and Rake, 2014; Knoben et al., 2006; Li et al., 2013; Zaheer and Soda, 2009). These agents form social relationships, such as formal and informal collaborations with each

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other, because they need to bring together diverse resources, knowledge, ideas and information embodied in others who can effectively and efficiently participate in a process that yields innovative outputs (Phelps et al., 2012). These relationships serve as social capital and represent the flowing and searching channels of knowledge and information (Adler and Kwon, 2002; Gonzalez-Brambila et al., 2013; Moran, 2005). These relationships also represent a lens through which social actors can evaluate each other and their knowledge stocks (Phelps et al., 2012). Ongoing debates exist on how social networks facilitate or constrain innovation outcomes of various agents. These debates mainly focus on the structural and relational characteristics of social networks and have been conducted across multiple research fields and multiple levels of analysis (Gonzalez-Brambila et al., 2013; Karamanos, 2012; Phelps et al., 2012; Uzzi and Spiro, 2005). For instance, at the inter-organizational level, sociologists as well as strategy and organizational behavior researchers have investigated how collaboration networks affect innovation outcomes of organizations. The key questions in these studies are whether ego-networks should be sparse or dense, whether organizations should bridge structural holes, or whether ties between organizations should be redundant or non-redundant (Adler and Kwon, 2002; Burt, 1992; Coleman, 1988; McFadyen et al., 2009; Rost, 2011). However, few studies have focused on the effects of this type of network embeddedness on organizations' technological innovation performances in terms of exploration and exploitation.

Knowledge is the core resource of organizations to achieve competitive advantage (Grant, 1996; Moorthy and Polley, 2010). Many scholars have regarded the knowledge base of an organization as a simple aggregation of its knowledge elements. Previous studies mainly focused on how the quantitative characteristics of organizations knowledge base influence their innovation outcomes (Ahuja and Katila, 2001; Boh et al., 2014; Carnabuci and Operti, 2013; Phelps, 2010; Quintana-García and Benavides-Velasco, 2008). Knowledge size, depth and diversity are the important characteristics of organizational knowledge base. The size or breadth of knowledge base of the organization has shown positive effects on its innovation outcomes (Ahuja and Katila, 2001; Boh et al., 2014). Moreover, technological knowledge diversification is found to have a stronger effect on exploratory innovative capability than on exploitative innovative capability (Quintana-García and Benavides-Velasco, 2008). Furthermore, knowledge diversity affects the association between collaborative integration and firms' abilities to innovate by both recombinant reuse and recombination creation (Carnabuci and Operti, 2013; Strumsky et al., 2011).

In contrast to the dominant focus on organizations' knowledge base described above, an epochal study by Yayavaram and Ahuja (2008) examined the structural aspects of organizations' knowledge base. They viewed a firm's knowledge base as a network formed by the coupling relationships among knowledge elements. These relationships record the past combination and affiliation of knowledge elements in the process of innovation, and then serve as the flowing and searching channels for knowledge and guide for future potential combination or recombination of knowledge elements (Phelps et al., 2012; Yayavaram and Ahuja, 2008). The study by Yayavaram and Ahuja (2008) on the worldwide semiconductor industry proved that the level of the knowledge structure decomposability influences invention-related outcomes of organizations. Moreover, Wang et al. (2014) integrated intra-organizational individual collaboration network and knowledge network into a research framework for the first time and found that the structural features of these two networks are distinct and affect individuals' exploratory innovation in different ways through separate mechanisms.

Except for the ground-breaking study by Yayavaram and Ahuja (2008) and Wang et al. (2014), no other systematic study on knowledge network is currently available. Existing studies blur the role of knowledge network embeddedness on innovation, which warrants further research. In contrast to research on simple aggregation of knowledge elements, the structural aspect of knowledge base is promising. Therefore, on the basis of extant research on the effects of social and knowledge networks on innovation, we aim to explore the structural characteristics of knowledge network and inter-organizational technology-based collaboration network in the emerging nano-energy field. We also aim to examine the possible effects of three network features, direct ties, indirect ties and non-redundancy among ties on organizational innovation in terms of exploitation and exploration.

2. Theoretical background and hypotheses

Since March's seminal work on the relationship between exploitation of old certainties and exploration of new possibilities in organization learning (March, 1991), many scholars have investigated exploitative and exploratory innovations (Gilsing and Nooteboom, 2006; Lavie et al., 2010; Quintana-García and Benavides-Velasco, 2008; Wang and Hsu, 2014). Exploitative innovation is characterized by a process of intensive search involving activities along organizations' existing knowledge dimension (Enkel and Gassmann, 2010; Lavie et al., 2010; March, 1991). Therefore, exploitative innovation improves the existing methods or materials used by organizations to achieve their goals. By contrast, exploratory innovation is rooted in an extensive search process pursuing opportunity areas that are new to organizations (Gilsing and Nooteboom, 2006; Lavie et al., 2010; March, 1991). Therefore, exploratory innovation brings in new methods or materials derived from knowledge dimension novel to a given organization (Karamanos, 2012; March, 1991; Wang and Hsu, 2014). In general, exploitative innovation deepens an organization's core knowledge base, whereas exploratory innovation broadens its existing knowledge base. In this study, exploitative and exploratory innovations have specific meanings within the context of technological invention: exploitative innovation means that a focal organization yields intimately familiar technological inventions, whereas exploratory innovation means that a focal organization yields technological novel technological inventions (Vanhaverbeke et al., 2006).

For exploitative and exploratory innovations, technology-based collaboration networks play important roles (Vanhaverbeke et al., 2006). In embedding in this type of network, organizations are given social capital that enables and constrains them to access, search for, transfer and apply related knowledge and information held by other actors in the network in the innovation process (Bierly et al., 2009; Vanhaverbeke et al., 2006). According to the extant studies at the inventor level, knowledge base structure of inventors in an intra-organizational knowledge network implies combinatorial opportunities, potentials and constraints arising from interactions among knowledge elements in past innovations (Wang et al., 2014; Yayavaram and Ahuja, 2008). Innovation builds on the existing knowledge base, and it is also characterized as a combinatorial or recombinatorial process of existing knowledge elements (Fleming, 2001; Schumpeter, 1934), even though we cannot rule out some ideas or inventions emerging with few antecedents (Arthur, 2009; Podolny and Stuart, 1995).

In this study, collaboration networks highlight the importance of social-based search, and knowledge networks highlight the importance of knowledge-based search. Three aspects of both networks probably connect with their benefits and constraints on exploitative and exploratory innovations. (1) First is the number of direct ties maintained by a focal organization in a collaboration network, and also the number of direct ties maintained by the knowledge elements of the focal organization in a knowledge network. A node's direct ties are the number of its unique direct connectors in a network (i.e., the size of its ego-network) (Ahuja, 2000; Gonzalez-Brambila et al., 2013). (2) Second is the number of indirect ties maintained by a focal organization, that is, the number of partners that the organization can reach in a collaboration network by its partners and their partners. We also focus on the number of indirect ties maintained by the knowledge elements of the focal organization in a knowledge network, which has a similar meaning as indirect ties in collaboration networks. A node's indirect ties reflect its reachability in a network (Ahuja, 2000; Bian, 1997). (3) Third is the degree to which an organization's direct partners connect to one another, that is, the extent of non-redundancy among ties. We employed network efficiency as the ratio of non-redundant links to total links of the organization's ego-network to measure non-redundancy (Ahuja, 2000; Burt, 1992). This network index captures whether structural holes exist in the organization's ego network (Burt, 1992). We also focus on the non-redundancy among ties in knowledge networks, used to capture the position of structural hole of knowledge elements.

2.1. Direct ties in a knowledge network

The number of direct ties of a knowledge element in a knowledge network (i.e., the number of other knowledge elements that it combined in previous technological inventions) indicates its combinatorial potential with other knowledge elements (Wang et al., 2014), whether the novel ones it has not combined with or the old ones it had already combined with in the past. Therefore, the number of direct ties of an organization's knowledge elements in a knowledge network is expected to be relevant in connection with the exploitative and exploratory innovations of it. Wang et al. (2014) contended that a knowledge element's combinatorial potential is affected by three aspects: (1) the subject matter's natural relatedness of the knowledge element with other ones (Quatraro, 2010). When seeking combinatorial opportunities, an organization is likely to search for subject-related knowledge elements through its own existing knowledge elements (Katila and Ahuja, 2002). (2) The combinatorial potential of a knowledge element is socially constructed, building on the strength of beliefs in the feasibility and desirability of combining it with other knowledge elements (Kuhn, 2012). The stronger the belief is, the more the resources will be allocated to it for seeking combinatorial opportunities. (3) A full understanding of the combinatorial experiences of the knowledge element in the past will increase its combinatorial potential in the future (Boh et al., 2014). Because, when seeking combinatorial opportunities for familiar knowledge elements, routines, standards and modules are formed, thus increasing the inventive efficiency (Levitt and March, 1988). The number of direct ties of a knowledge element may be characterized as a comprehensive reflection of the given three aspects (Wang et al., 2014).

Therefore, when direct ties of an organization's knowledge elements are too few, the organization tends to have low combinatorial potentials. In this case, the organization needs to spend great efforts and expensive learning costs to explore combinatorial opportunities for its knowledge elements with its existing or novel ones. Moreover, few direct ties may reduce the likelihood of success in combinations, thus reducing innovation performance. However, when the direct ties of an organization's knowledge elements are too many, combinatorial potentials are probably low as well, as the combinatorial potential of any knowledge element has an upper limit at which its value is exhausted (Wang et al., 2014). When a knowledge element eventually reaches this point, its further combination with other knowledge elements ceases to be fruitful. Therefore, too many direct ties also have an adverse effect on innovation performance. Therefore, we expect the following hypotheses:

Hypothesis 1a. The number of direct ties of knowledge elements of an organization in a knowledge network is expected to have an inverted U-shaped effect on its exploitative innovation.

Hypothesis 1b. The number of direct ties of knowledge elements of an organization in a knowledge network is expected to have an inverted U-shaped effect on its exploratory innovation.

2.2. Direct ties in a collaboration network

Technological collaboration with direct partners provides information benefits (Ahuja, 2000; Gonzalez-Brambila et al., 2013; Vanhaverbeke et al., 2006; Wang et al., 2014). Wang et al. (2014) presented three major types of information acquired from cooperative research at the individual level: (1) information related to the distribution of knowledge among inventors, (2) information pertaining to the latest developments and trends of research and (3) information bearing on the inventors and the interactions among them. Moreover, technological collaboration with direct partners also leads the reduction in costs and risks involved in the process of innovation (Ahuja, 2000; Vanhaverbeke et al., 2006). On the one hand, the new outcomes are available to all the partners that participate in the cooperating innovation (Wang et al., 2014). On the other hand, collaboration with direct partners facilitates the application of complementary skills and knowledge owned by other organizations (Vanhaverbeke et al., 2006). Therefore, each partner can potentially receive more innovation performance in collaborative activities than in stand-alone ones.

However, technological collaboration with direct partners poses threats as well (Demirkan et al., 2013; Vanhaverbeke et al., 2006). Collaboration means joint efforts to overcome challenges in the process of innovation, and partners share and exchange proprietary information and resources in this process (Demirkan and Demirkan, 2012). Therefore, collaboration is very likely to entail the risks of free-riding behaviors or unexpected spillovers (Wu, 2008), and these adverse aspects have a tendency to grow with the number of partners. Moreover, organizations may be subject to information overload and diseconomies of scale when the number of their partners exceeds a certain number (Gulati et al., 2012). In this case, organizations just have to spend more time and energy to monitor and manage their collaborative relationships. Therefore, we expect the following hypotheses:

Hypothesis 2a. An organization's number of direct ties in a collaboration network is expected to have an inverted U-shaped effect on its exploitative innovation.

Hypothesis 2b. An organization's number of direct ties in a collaboration network is expected to have an inverted U-shaped effect on its exploratory innovation.

2.3. Indirect ties in a knowledge network

Indirect ties of a knowledge element refer to the knowledge elements that it can reach through its connectors and their connectors. In general, the more the indirect ties of a knowledge element are, the more the other knowledge elements can be searched for by it. Searching and understanding previous combinations through indirect ties may promote recombination or new combination.

On the one hand, indirect ties of a focal organization's knowledge elements may deliver abundantly recombinational or combinatorial opportunities, as knowledge linkage relationships record the contents and experiences of knowledge couplings in previous inventions (Carnabuci and Bruggeman, 2009; Fleming, 2001; Yayavaram and Ahuja, 2008). Regarding to a focal organizational, recombination means reusing and refining its known knowledge combinations to develop new applications or solve new problems, which is useful for the organization's exploitative innovation; while, new combination means employing never combined knowledge elements of the organization that include its existing ones and the ones novel to it, combining with existing ones is useful for exploitative innovation and combining with novel ones is useful for exploratory innovation. On the other hand, searching by indirect ties may promote interactions of a number of diverse ideas and routines. These collisions may generate novel ideas, which are very useful for exploratory innovation. Therefore, we provide the following hypotheses:

Hypothesis 3a. The number of indirect ties of an organization's knowledge elements in a knowledge network favors its exploitative innovation.

Hypothesis 3b. The number of indirect ties of an organization's knowledge elements in a knowledge network favors its exploratory innovation.

2.4. Indirect ties in a collaboration network

Collaborative ties also can be served as channels for information between focal organizations and their indirect partners (Ahuja, 2000; Karamanos, 2012; Vanhaverbeke et al., 2006). Regarding to a focal organization, its indirect partners refer to the ones that it can reach through its partners and their partners. Therefore, organizations can acquire information both from its direct and indirect partners in collaboration networks. In the sense, indirect ties serve as a conduit of information, thus they may entail some benefits for both exploitative and exploratory innovation in some degree.

Even if indirect ties can bring in information for organizations, they may also contain disadvantages as well. Indirect ties are inclined to deliver novel information on unfamiliar domains (Granovetter, 1973), which seems to be useful for exploitative and exploratory innovation, especially the latter, because exploration implies that organizations departs from their existing knowledge dimension (March, 1991). However, information gained from indirect partners may be distorted during its transferring process (Vanhaverbeke et al., 2006). Therefore, some real information may never reach a focal organization, because actors readily understand the information they receive in their own ways that lead to misinterpretation or misunderstanding (Hansen, 2002). While, detecting and eliminating noisy information is costly (Kogut and Zander, 1993). This distorted information is bad for exploitative and exploratory innovation. In addition, indirect ties deliver little specific and fine-grained information which is more useful for exploitation innovation (Gilsing and Nooteboom, 2006). Furthermore, information can also be channeled in the opposite direction, that is, from focal organizations to their indirect partners (Gulati and Gargiulo, 1999). This reverse diffusion may readily cause unintended spillovers (Vanhaverbeke et al., 2006), which are extremely difficult to monitor or enforce sanctions.

Given the discussions above, we expect the disadvantages from indirect ties to offset their benefits and give the following hypotheses:

Hypothesis 4a. The number of indirect ties of an organization in a collaboration network hinders its exploitative innovation.

Hypothesis 4b. The number of indirect ties of an organization in a collaboration network hinders its exploratory innovation.

2.5. Non-redundancy among ties in a knowledge network

An investigation on the effects of non-redundancy among ties of an organization's knowledge elements in a knowledge network on exploitative and exploratory innovations is ongoing. The egocentric network of a knowledge element is a non-redundant or sparse structure if the knowledge element has been combined in previous inventions with other knowledge elements that are themselves disconnected from each other (Carnabuci and Bruggeman, 2009; Wang et al., 2014), with the redundancy among the alters being minimal (Burt, 1992, 2004). An organization with many non-redundant knowledge network structures has few constraints in exploring new ideas because it is rarely affected by knowledge inertia, which is a common phenomenon in redundant network structures (Cheon et al., 2014; Gargiulo and Benassi, 2000). Herein, inertia means the tendency in which substitution or change is passive because of the redundancy network structures (Cheon et al., 2014; Kim et al., 2006). Therefore, a redundant knowledge network structure may have a negative effect on exploring new ideas because of the following reasons. Firstly, organizations prefer to perform local search around the vicinities of their knowledge elements' ego-networks. However, in redundant network structures, organizations are vulnerable to the inertial tendencies of local search. Secondly, exploring novel ideas is associated with searching, learning and opportunity costs (Wang et al., 2014). Given these costs, when an organization is locked in redundant knowledge structures, it is possibly reluctant to invest time and money to explore new ones, but not reluctant to focus on refining and recombining extant knowledge elements. Thirdly, exploring new ideas is full of unpredictable, uncertainty and emergency (March, 1991; Wang et al., 2014) and they enhance the inertial behaviors of organizations (Benner and Tushman, 2003). Redundant ties deliver many contents and experiences of the preceding couplings of knowledge elements, which are helpful for organizations to understand, assimilate and apply these knowledge elements in the future. Therefore, in redundant knowledge structures, organizations may prefer to focus more on the pursuit of utilization of extant knowledge elements rather than explore new ideas. Therefore, we expect the following hypotheses:

Hypothesis 5a. Non-redundancy among ties of an organization's knowledge elements in a knowledge network hinders its exploitative innovation.

Hypothesis 5b. Non-redundancy among ties of an organization's knowledge elements in a knowledge network favors its exploratory innovation.

2.6. Non-redundancy among ties in a collaboration network

When a focal organization's partners do not link to one another, they form a non-redundant network structure, thus minimizing the redundancy between them (Burt, 1992, 2004). The organization can have access to a number of novel information and ideas in a non-redundant network structure (Burt, 2004), such as the latest developments and research trends, the distribution of knowledge or skills among organizations and the features related to its partners and their interactions (Wang et al., 2014). Timeliness is a remarkable feature of information obtained by an organization through its disconnected partners (Burt, 1992). Therefore, an organization whose ego-network is rich in non-redundant ties can capture emerging opportunities and threats and can also learn about possible new partners quickly. These capabilities may be beneficial for both exploitative and exploratory innovations. Moreover, an organization whose ego-network is rich in non-redundant ties may benefit from structural autonomy in innovation activities (Burt, 1992, 2004). Therefore, it can avoid restrictions caused by interactions among its partners (Cheon et al., 2014), such as opinions of leaders and some prevailing cognitive schemes that are likely to occur in redundant network structures (Karamanos, 2012; Vanhaverbeke et al., 2006). Therefore, we expect the following hypotheses:

Hypothesis 6a. Non-redundancy among ties of an organization in a collaboration network favors its exploitative innovation.

Hypothesis 6b. Non-redundancy among ties of an organization in a collaboration network favors its exploratory innovation.

3. Data, variables and method

3.1. Data, knowledge representation and network construction

We tested our hypotheses in the context of the nano-energy field, which emerges from the application of nanotechnology in the energy sector (Diallo et al., 2013; Tegart, 2009). This field is highly dynamic, and it is also characterized by advances in science and technology and a high level of interactions among multi-disciplinary and multi-technological fields (Guan and Liu, 2014, 2015; Menéndez-Manjón et al., 2011). Therefore, numerous knowledge stocks and many events of exploitative and exploratory innovations can be observed. Our main dependent and independent variables are calculated based on nano-energy patent data granted to organizations. Although, patent-based indicators have certain limitations, a large body of research has demonstrated their validity as proxies to measure innovative activities (Belenzon and Patacconi, 2013; Griliches, 1998; Kleinknecht and Reinders, 2012). Moreover, all the examinations in this study are limited to a single field—the nano-energy field. Therefore, the different propensities in patenting across industries, which commonly occur in using patent-based indicators (Mansfield, 1986), are not an issue in this research.

We extracted the nano-energy patents data from the frequently used Derwent Innovation Index database (DII). We chose to use this database because it is the most comprehensive patent database worldwide covering patent information issued by more than 100 countries and 40 patent authorities, including USPTO, EPO, JPO and SIPO and so on. Therefore, nano-energy patents extracted from this database can reflect the developmental state of global technology in this field. Moreover, this database offers the descriptive title and abstract for each patent in detail, which are rewritten by technical experts coming from various fields using colloquial and concise English language. This point is very helpful to distinguish whether a patent extracted from DII belongs to our concerned nano-energy field.

To accurately identify and capture nano-energy patents from the DII database, we closely followed the keywords searching strategy utilized by Menéndez-Manjón et al. (2011), Guan and Liu (2014, 2015) and Liu and Guan (2015). That is, keywords relating to nanotechnology and energy technology included in the titles or abstracts of patents are used to search and select nano-energy patents. Two steps were performed to obtain nano-energy patents. Firstly, we searched patents in the nanotechnology field and the energy field, respectively. Then we performed Boolean logic “AND” operation on the search results of the nanotechnology patents and the energy patents obtained in the first step to acquire nano-energy patents. The searching terms adopted are defined in Appendix 1 (Guan and Liu, 2014, 2015; Liu and Guan, 2015).

We performed the retrieving processes in May 2014. To eliminate non-nano-energy patents, we carefully checked the frontpage of each patent. After a thorough cleaning process, we finally identified 40,502 nano-energy patents granted during the time period of 1991–2013 among which there are 36,521 nano-energy patents whose assignees are firms, universities or institutes. Fig. 1 shows the dynamic evolution of the nano-energy patent granted each year. As indicated by this figure, very few nano-energy were granted throughout the 1990s, but a considerable growth was observed during the 2000s. Therefore, we conducted our research exclusively based on the nano-energy patents granted in the last 14 years (2000–2013), unless otherwise stated.

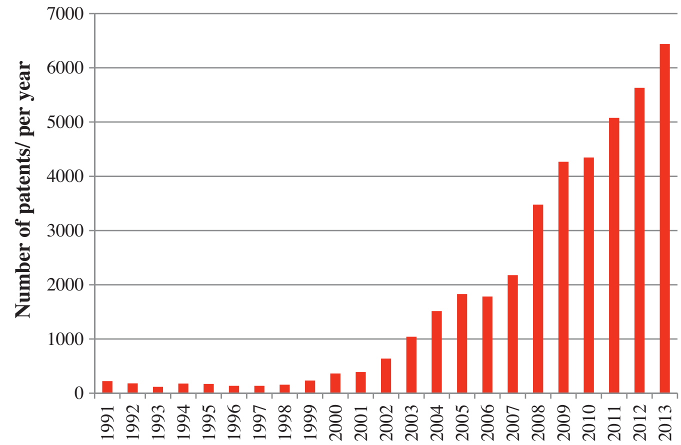

Fig. 1. The number of nano-energy patent granted per year, 1991–2013.

Technology codes in patents involving international patent classification codes (IPC) defined by the World Intellectual Property Organization (WIPO) or technological classes defined by the United States Patent and Trademark Office (USPTO) have been considered to be valid proxies for knowledge elements or components (Carnabuci and Bruggeman, 2009; Carnabuci and Operti, 2013; Dibiaggio et al., 2014; Guan and Liu, 2015; vom Stein et al., 2014). Considering the availability of data, we took IPC codes as proxies of knowledge elements. By adopting a hierarchical structure, the IPC classification system categorizes any a patent into a section, class, subclass, main group and subgroup. However, many studies exploited a four-digit IPC code to denote a knowledge element, which includes a section, class and subclass (Guan and Liu, 2015; Park and Yoon, 2014). It is mainly because four-digit IPC codes actually can sufficiently express the technological or knowledge features of a patent (Guan and Liu, 2015). Following the common practice, four-digit IPC codes are also exploited in this paper to denote the knowledge elements in the nano-energy field. Each nano-energy patent can be categorized into one or more four-digit IPC codes after we extracted and eliminated duplicated the duplicates.

A patent may involve one or multiple assignees that are organizations and may also involve one or more four-digit IPC codes. We constructed a technology-based collaboration and knowledge networks in the nano-energy field by considering this information. Collaboration networks are constructed based on joint assignees of each patent, and knowledge networks are constructed based on the co-application of four-digit IPC codes in each patent. We adopted a commonly used five-year moving window to construct them with the help of the Science of Science (Sci^2) Tool software.

Because of renaming phenomena or spelling mistakes, a same organization may have several different names. Thus, we must have to harmonize the names of organizations before we built inter-organizational collaboration networks. To achieve this, we firstly extracted the full names of organizations from nano-energy patents, then searched their standard names from their official websites and the World-Wide-Web and finally unified the different names of a same organization into a standard name. In the process of unification, we also identified affiliated organizations and merged them into the parent organizations.

Our empirical sample consists of Firms, Universities and Institutes that were granted at least one nano-energy patent both during the previous five years and in the subsequent observation year (i.e., during the $t-1$ to $t-5$ year and in year t). In accordance with this criterion, we finally identified 919 innovative organizations located in North America, Europe and Asia and 5107 observations during 2000–2013. Therefore, our empirical sample was composed of unbalanced panel data.

3.2. Variables

3.2.1. Dependent variables

Despite some obvious limitations, patent-based measures are widely accepted as proxies of innovative outputs (Archibugi, 1992; Griliches, 1998; Schmookler, 1966). In this study, two dependent variables, namely, exploitative and exploratory innovations are also developed based on each organization's explorative and exploratory patents, respectively. A challenge in using patents as indicators of innovative outputs is that patent values have highly skewed distribution, and most of the patents rest at the very tail of the value distribution (Gambardella et al., 2008; Gittelman, 2008; MingJi and Ping, 2014). Therefore, patent counts weighted by forward citations (Trajtenberg, 1990), patent family size (Putnam, 1996), years of renewal (Schankerman and Pakes, 1986) and the number of claims in patent application (Tong and Frame, 1994) are used and verified to be better measures of patent value. To prevent the limitations of simple patent counts, we followed Guan and Zhao (2013) and used patent family size as a value criterion after we considered the availability of our data. Patent family size is the number of jurisdictions in which patent protection for a single invention has been filed (Putnam, 1996). Family size is significantly correlated with patent value, especially the economic value (Fischer and Leidinger, 2014; Harhoff et al., 2003). Seeking patent protection for a single invention in multiple patent authorities involves additional costs (e.g., patent attorney, examination and translation fees) (Fischer and Leidinger, 2014; Putnam, 1996). If an applicant chooses to extend the protection range of an invention, then the benefits acquired from the protection should exceed the extra costs. Therefore, we used two alternative operationalizations to increase the robustness of our analyses: 'weighted patent counts' (each exploitative and exploratory patent weighted by its patent family size) and 'simple patent counts' (the number of exploitative and exploratory patents). In this study, patent family size is measured by the number of patent authorities at which a given invention has been protected (Guan and Zhao, 2013; Squicciarini et al., 2013).

To calculate our dependent variables, we first identified that each nano-energy patent granted to a focal organization in a given observation year could be classified into an exploitative or exploratory patent based on the technological classes used to instantiate them. To achieve this aim, we closely followed the approach adopted by Vanhaverbeke et al. (2006) and Gilsing et al. (2008) and determined which technology classes of an organization are exploitative and exploratory in the observation year. Thus, we calculated the technology profiles of a focal organization in the previous five years and in the subsequent observation year, respectively. Then, we matched the technology profiles of each organization in these two time periods to distinguish exploitative and exploratory technology classes. The technology profiles of a focal organization are produced by adding up all its technology classes (i.e., four-digit IPC codes). A technology class is labeled as exploitative if it is present in the organization's stock of technology classes both during the preceding five years and in the subsequent observation year (Gilsing et al., 2008; Vanhaverbeke et al., 2006). By contrast, a technology class is viewed as exploratory if it is present in the organization's stock of technology classes in the observation year, but absent during the preceding five years (Gilsing et al., 2008; Vanhaverbeke et al., 2006). A nano-energy patent of an organization is viewed as an exploitative patent, if the patent is instantiated by all the exploitative technology classes; a patent is viewed as an exploratory patent, if the patent is instantiated by at least one exploratory technology class.

Based on the above discussions, the exploitative innovation performance of a focal organization in an observation year t can be determined by:

$$\text{Exploitative Innovation} = \sum_{i=1}^{m} S_i \tag{1}$$

where m denotes the number of exploitative patents granted to the organization in the observation year t, and S_i represents the family size of the patent i. Likewise, the exploratory innovation performance of a focal organization in the observation year t can be determined by:

$$\text{Exploratory Innovation} = \sum_{j=1}^{n} S_j \tag{2}$$

where n denotes the number of exploratory patents granted to the organization in the observation year t, and S_j is the family size of the patent j.

3.2.2. Independent variables

The independent variables in this study are direct ties, indirect ties and the non-redundancy among ties in both collaboration and knowledge networks. We calculated these variables using UCINET 6.516. An organization generally possesses multiple knowledge elements. Therefore, to work out an independent variable of an organization's knowledge elements in a knowledge network, we adopted the approach of Wang et al. (2014) and averaged the scores of all the knowledge elements of an organization in the knowledge network.

3.2.3. Mean knowledge direct ties

This variable indicates the average number of direct ties of an organization's knowledge elements in a knowledge network. This value is calculated through the following three steps. Firstly, we calculated how many knowledge elements are directly connected to each knowledge element (i.e., the size of each knowledge element's ego-network) in the knowledge network. Secondly, we identified the knowledge portfolio of the organization by aggregating all of its knowledge elements during a five-year period prior to an observation year t. Thirdly, we averaged the number of direct ties of each knowledge element of the organization. To test the inverted U-shaped effect of this variable on innovation performance, we also introduced its squared term.

3.2.4. Mean knowledge indirect ties

This variable indicates the average extent to which an organization's knowledge elements can indirectly connect to other knowledge elements in a knowledge network. We took this variable as an average value of the number of indirect ties of each knowledge element of the organization. The number of a knowledge element's indirect ties is defined as a distance weighted centrality by attaching declining weights in the form of $1/k$ to its indirect ties (Borgatti, 2003). It can be obtained by using the following formula:

$$\text{Indirect ties} = \sum_{k=2}^{m} \frac{N_k}{k} \tag{3}$$

where k denotes the number of steps through which a focal knowledge element can indirectly reach other elements (i.e., the path distance), m refers to the maximum step that the focal knowledge

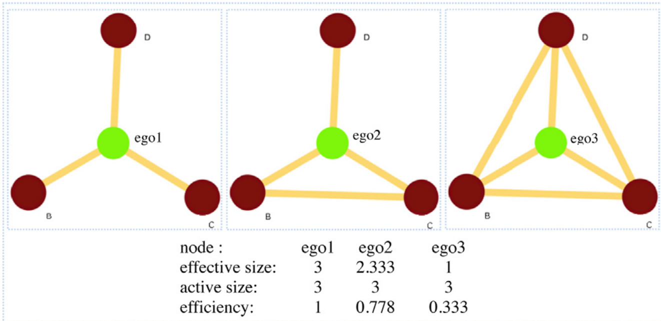

Fig. 2. Examples for efficient and inefficient networks.

element can reach the farthest ones, and N_k expresses the number of ties the focal knowledge element can reach in the step of k.

3.2.5. Mean knowledge network efficiency

The value of this variable can be calculated by averaging the network efficiency scores of each knowledge element of an organization. We chose network efficiency as a measure of non-redundancy among ties in knowledge networks (Burt, 1992). Network efficiency of a knowledge element in a knowledge network indicates the average degree to which its direct connecters are not connected to each other (Burt, 1992). It can be obtained through dividing the effective size of its ego-network by the actual size. The network efficiency of a knowledge element i in a knowledge network can be computed by the following formula:

$$\text{Network efficiency} = \frac{\sum_j (1 - \sum_q p_{iq} p_{jq})}{C_i}, \quad q \neq i, j \tag{4}$$

where p_{iq} is the proportion of relations that the knowledge element i invested in the connection with the knowledge element q, p_{jq} has a similar meaning, $\sum_q p_{iq} p_{jq}$ denotes the redundancy of ties between the knowledge element i and the knowledge element j, the section of numerator refers to the effective size of i's ego-network and C_i expresses the total number of its direct connecters (i.e., its active size) (Burt, 1992). A higher score in network efficiency reflects that the knowledge element i's ego-network is rich in structural holes. That is, its direct connecters are not connected to each other. If no redundant ties exist, then the network efficiency is 1. To illustrate network efficiency clearly, simple network diagrams in Fig. 2 contrast on efficient and inefficient networks. As displayed by this figure, the green node's network efficiency decreases with the increases of the number of redundant ties.

3.2.6. Direct ties in a collaboration network

This variable is defined as the number of partners with whom an organization is directly connected in a collaboration network (i.e., the size of the organization's ego-network). We also introduced the squared term of this variable to test its inverted U-shaped effect on innovation performance.

3.2.7. Indirect ties in a collaboration network

This variable is the number of partners that an organization can indirectly reach through its partners and their partners in a collaboration network. We computed the value of this variable by using Formula (3), where, k is the number of steps through which a focal organization can reach its partners, m is the maximum step that it can reach the farthest ones, and N_k is the number of its k-step-distant partners.

3.2.8. Collaboration network efficiency

We also chose network efficiency as a measure of the non-redundancy among ties in collaboration networks. It can be calculated by using Formula (4), where, p_{iq} is the proportion of collaborative relations that the organization i invests in the organization q, p_{jq} has a similar meaning, and C_i is the number of direct partners of the organization i. If the organization i has a higher value in its network efficiency, then it occupies a position that is full of structural holes (Burt, 1992). If all of the organization i's direct partners are not connected with each other, then the network efficiency is 1, which indicates that the redundancy of ties among its direct partners is at minimum degree.

Comparing to the measurements of the non-redundancy among ties, the measurements of direct ties and indirect ties both have a different scale. Thus, we normalized them using the method of min-max normalization before we performed our regression estimations.

3.2.9. Control variables

Although we focused on the effects of embeddednesses in knowledge and collaboration networks on exploitative and exploratory innovations, other factors may also affect them. Therefore, we introduced additional control variables, namely, dummy variables, R&D intensity and exploitative and exploratory learning abilities. Extant studies have presented that organizations in different countries have a unique propensity to apply for patents (Archibugi, 1992; Arundel and Kabla, 1998). Therefore, dummy variables are introduced to control changes over countries in patenting propensity. Organizations in our sample are located in North America, Europe or Asia, and thus we included two dummy variables. The default value is North America. Different types of organizations may also have different propensities to apply for patents. Thus, we included dummy variables to indicate that an organization is a University, an Institute or a Firm (the default is Firm). Finally, we included annual dummy variables to control changes over time in patenting propensity.

3.2.10. R&D intensity

R&D expenditures are not accessible for most organizations, especially those in nano-energy research. To ensure the robustness of our research results, we followed the common practice and included patent stocks of a focal organization in the past four years as its proxy (Gonzalez-Brambila et al., 2013; Guan and Zhao, 2013; Schilling and Phelps, 2007). Patent stocks have been demonstrated to be highly correlated with annual R&D expenditures (Schilling and Phelps, 2007).

3.2.11. Exploitative and exploratory learning ability

Exploitative and exploratory innovations are not separate events, but are mutually related to and build on each other to some extent, whether positively or negatively (Gupta et al., 2006; Vanhaverbeke et al., 2006). To control for this possible effect, we included the number of lagged exploitative and exploratory patents of a focal organization in the year before the observation year t as proxies for exploitative and exploratory learning abilities.

As indicated above, we summarized the definitions of distinct variables in Table 1.

3.3. Model specification

Our dependent variables are count variables and take only non negative values. Therefore, we should adopt either Poisson models or negative binomial models. For the pooled cross-section data of our sample, two dependent variables both have a high extent of variance versus their own means, thus, we chose to employ negative binomial regression models to correct for overdispersion (Barnett, 1997). In addition, we also grips with the choice between standard negative binomial models and zero-inflated negative binomial models. Zero-inflated negative binomial models can

Table 1
Definitions of distinct variables.

Variable name	Variable description
Dependent variables	
Exploitative innovation	The family size-weighted exploitative patents granted to a focal organization in a given year *t* categorized by technology classes that are all used to instantiate the organization's patents during the previous five years.
Exploratory innovation	The family size-weighted exploratory patents granted to a focal organization in a given year *t* categorized by technology classes with at least one in each patent that has not been used to instantiate the organization's patents during the previous five years.
Independent variables	
Mean knowledge direct ties	The mean number of knowledge elements to whom a focal organization's knowledge elements are directly connected in a knowledge network (standardized variable).
Mean knowledge indirect ties	The mean value of distance weighted centrality: the mean number of knowledge elements to whom a focal organization's knowledge elements are indirectly connected in a knowledge network weighted by declining step-distances (standardized variable).
Mean knowledge network efficiency	The mean level of non-redundancy among ties of a focal organization's knowledge elements in a knowledge network.
Direct ties in collaboration network	The number of partners to whom a focal organization is directly connected (standardized variable).
Indirect ties in collaboration network	Number of partners to whom a focal organization is indirectly connected and weighted by declining step-distances (standardized variable).
Collaboration network efficiency	The non-redundancy among ties in a focal organization's ego-network is measured by network efficiency which is gotten through dividing the effective size of by the actual size of its ego-network.
Control variables	
Asia	Dummy variable is set to one if a focal organization is in Asia (North America is the default value).
Europe	Dummy variable is set to one if a focal organization is in Europe (North America is the default value).
University	Dummy variable is set to one if a focal organization is a university (Firm is the default value).
Institute	Dummy variable is set to one if a focal organization is an institute (Firm is the default value).
Year	Dummy variable is used to indicate a particular year (2005–2013, 2013 is the default value).
R&D intensity	The stock of patents granted to a focal organization in the past four years before a given year *t*.
Exploitative learning ability	The number of exploitative patents granted to a focal organization in the $t-1$ year.
Exploratory learning ability	The number of exploratory patents granted to a focal organization in the $t-1$ year.

account for the prevalence of zero outcomes in the data (Greene, 1994). Approximately 40% and 16% of our two dependent variables is equal to zero. We performed Vuong tests and the test results indicate neither model is preferred because of the Vuong statistics $|V| < 1.96$ (Vuong, 1989). Besides, the panel data option in STATA 12.0 is not available for zero-inflated negative binomial models. Considering above, we first reported the estimation results of standard negative binomial models that consider the grouping structure of our data. Then we check the robustness of our results using zero-inflated binomial models that do not consider the grouping structure of our data. Therefore, for standard negative binomial models, we examined random effects because that fixed effects models can produce biased estimates for panel data whose period is short (Greene, 2008; Hsiao, 1986).

4. Results

4.1. Features of collaboration and knowledge networks

The pattern of technological knowledge exchange and information diffusion among organizations in their innovative processes reflects the collaborative ties among them. In particular, one aspect of inter-organizational collaborative networks that directly affect the technological knowledge exchange and information diffusion among organizations is the degree of integration of the network structure (Carnabuci and Operti, 2013). The integration of a network structure means the extent to which it consists of one interconnected component that provides nodes with more opportunities to reach one other (Carnabuci and Operti, 2013). Likewise, the pattern of knowledge association or combination reflects the knowledge ties among knowledge elements. Similarly, the associative integration of knowledge networks affects knowledge combination and creation in the future (Yayavaram and Ahuja, 2008). Our results show that inter-organizational collaboration networks and knowledge networks in the nano-energy field differ in the degree of integration. Fig. 3a and b illustrates the integration of collaboration and knowledge networks in 2000–2004. Inter-organizational collaboration networks in the nano-energy field are characterized by a low degree of collaborative integration. This type of network is fragmented across many unconnected components, resulting in few opportunities for knowledge exchange, information diffusion and joint problem solving. Conversely, we found that knowledge networks in the nano-energy field feature as more associative integrated network structures, in which knowledge elements largely connect to one other by direct or indirect ties. Therefore, knowledge networks provide organizations with many opportunities for knowledge findings and redefinition of organizational tasks through continuous knowledge searching.

Moreover, our results demonstrate that inter-organizational collaboration networks and knowledge networks in the nano-energy field are decoupling. In other words, at an identical time period, an organization's positions in a collaboration network do not necessarily correspond to the positions of its knowledge elements in a knowledge network. We also took network characteristics of some organizations during the time period 2000–2004 as examples. Nanosolar Inc had no direct or indirect partners because it did not participate in inter-organizational collaboration research in this time period, but the average direct and indirect ties of its knowledge elements in the knowledge network are 96.56 and 135.19, and the average knowledge network efficiency of its knowledge elements is 0.83. Univ Tsinghua had 5 direct partners but no indirect partners in the inter-organizational collaboration network, and its collaboration network efficiency is 0.79; the average scores of these three indicators of its knowledge elements in the knowledge network are 103.94, 131.70 and 0.82. Hitachi Ltd had 14 direct partners and 34.39 indirect partners in the inter-organizational collaboration network, and its collaboration network efficiency is 0.85; while the average scores of these three indicators of its knowledge elements in the knowledge network are 97, 136.84 and 0.81.

Referring to Wang et al. (2014)'s examinations, this decoupling phenomenon can be explained from the following three aspects. Firstly, knowledge elements are not equally distributed across organizations. In general, most organizations possess more than a single knowledge element, and most knowledge elements are owned by more than one organization. Secondly, collaboration research between organizations may involve a single or multiple knowledge elements, and two knowledge elements may be combined by more than one organization. Thirdly, not all organizations

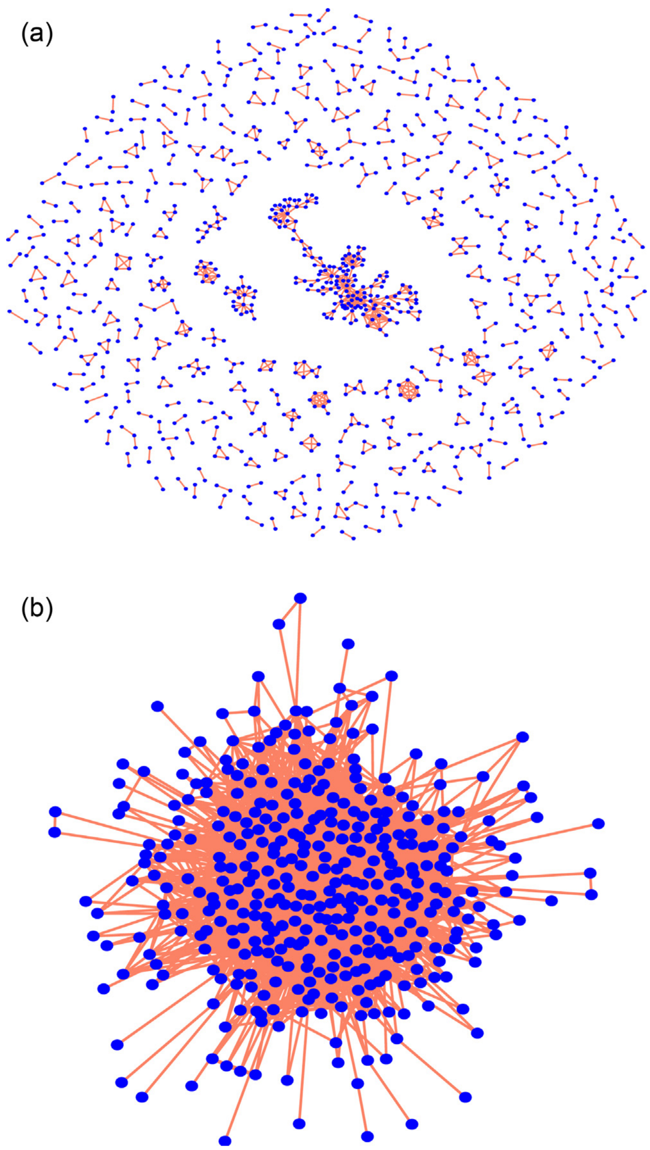

Fig. 3. (a) A technology-based collaboration network characterized by a low degree of collaborative integration, 2000–2004. (b) A knowledge network in the nano-energy field characterized by a high degree of associative integration, 2000–2004.

participate in inter-organizational collaborative research. An organization may not participate in inter-organizational collaboration research, but it possesses more than a single knowledge element. Therefore, the decoupling of collaboration and knowledge networks is a regular event rather than an exceptional one.

Fig. 4 provides a diagram to illustrate the decoupling of inter-organizational collaboration and knowledge networks in the nano-energy field. In this figure, a collaboration network is located in the top section, a knowledge network is in the bottom section, and the organizations' ownership of knowledge elements is in the middle section. Compared with organizations A and C, organization B, the degree centrality of which is 6 and network efficiency is 0.833, is located in a more central position and has less network constraint in the collaboration network. However, in the knowledge network, the average degree centrality of organization A's and C's knowledge elements are both higher than that of B (A = 7, B = 3.25, and C = 6.333). Moreover, the average value of knowledge network efficiency for organizations A, B and C is 0.567, 0.333 and 0.569, respectively. Therefore, organization A and C both have less knowledge network constraints than organization B.

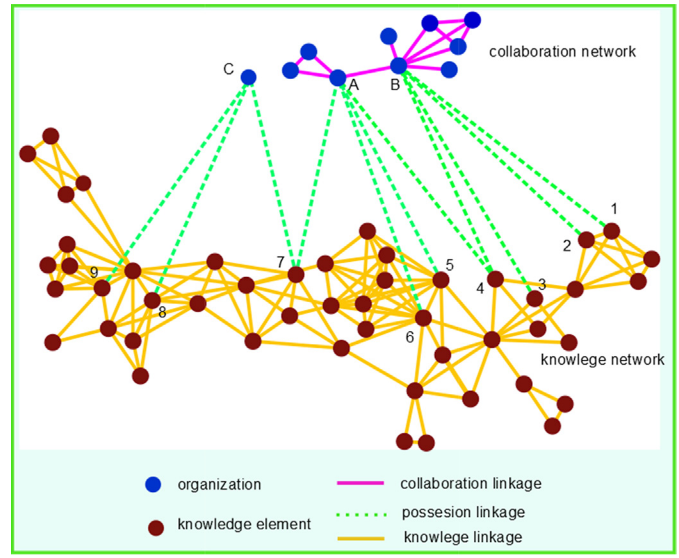

Fig. 4. An example for the decoupling of collaboration and knowledge networks (referring to Wang et al., 2014).

As shown in Table 2, the correlation levels between variables related to the knowledge networks and the matching variables pertaining to the collaboration networks are low. For instance, the correlation coefficient between the average knowledge network efficiency of an organization's knowledge elements and its collaboration network efficiency is −0.040. These considerably low correlations further confirm that collaboration and knowledge networks are loosely coupled. Therefore, an organization's positions in an inter-organizational collaboration network do not necessarily reflect those of its knowledge elements in a knowledge network.

4.2. Regression results

Table 2 presents the descriptive statistics of variables and their correlations. Multicollinearity does not seem to be a problem in this paper because the variance inflation factor of each variable is lower than 10, and the condition index for each of the variables is lower than 30. Table 3 reports the random-effects negative binomial panel regression results for exploitative and exploratory innovations. Models 1 and 5 both represent the basic model, which includes only our concerned control variables. Models 2 and 6 add independent variables pertaining to the knowledge network structures to Models 1 and 5, respectively. Models 3 and 7 add the variables related to the collaboration networks structure to Models 1 and 5, respectively. Models 4 and 8 both represent the full model, including all our concerned control and independent variables.

Hypothesis 1a predicts that the number of direct ties of knowledge elements of an organization in a knowledge network has an inverted U-shaped effect on its exploitative innovation. Models 2 and 4 in Table 3 both provide statistical support for this hypothesis at the significant level of $p < 0.01$. Therefore, a small number of direct ties of an organization's knowledge elements in a knowledge network are beneficial for the organization's exploitative innovation. With the increase in the number of direct ties up to a specific point, the consequences of diminishing returns begin to dominate. Hypothesis 1b forecasts the inverted U-shaped relationship of the number of direct ties of an organization's knowledge elements in a knowledge network with its exploratory innovation performance. According to the reported regression results of Models 6 and 8 in Table 3, this hypothesis is not confirmed. Therefore, the number of direct ties of an organization's knowledge elements in a knowledge network has different effects on the organization's exploitative and exploratory innovations.

Table 2
Descriptive statistics and correlation matrix.

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Exploitative innovation															
2. Exploratory innovation	.353**														
3. Mean knowledge direct ties	−.128**	−.193**													
4. Mean knowledge indirect ties	.205**	.032*	−.352**												
5. Mean knowledge network efficiency	−.098**	.037**	.380**	−.302**											
6. Direct ties in collaboration network	.424**	.204**	−.086**	.197**	−.080**										
7. Indirect ties in collaboration network	.237**	.021	.154**	.253**	−.057**	.528**									
8. Collaboration network efficiency	.124**	.076**	−.002	.152**	−.040**	.317**	.336**								
9. Asia	.036**	−.117**	.140**	−.050**	−.081**	.099**	.143**	.009							
10. Europe	−.017	.095**	−.051**	.020	.037**	.001	−.073**	.042**	−.512**						
11. University	−.098**	−.092**	.084**	.024	.055**	−.036*	−.053**	−.039**	.049**	−.089**					
12. Institute	.019	.035*	.011	.018	.011	.040**	−.015	.013	.061**	−.035*	−.238**				
13. R&D intensity	.725**	.262**	−.159**	.321**	−.149**	.571**	.340**	.188**	.110**	−.099**	−.070**	.037**			
14. Exploitative learning ability	.717**	.225**	−.098**	.256**	−.120**	.484**	.299**	.151**	.123**	−.088**	−.071**	.031*	.896**		
15. Exploratory learning ability	.406**	.313**	−.202**	.181**	−.118**	.308**	.119**	.153**	.075**	−.080**	0.022	.061**	.510**	.368**	
Mean	4.678	5.309	135.995	156.015	0.818	2.665	51.293	0.607	0.587	0.155	0.303	0.115	11.718	1.869	1.832
S.D.	11.585	7.435	35.031	20.823	0.021	4.226	77.389	0.418	0.492	0.362	0.460	0.319	17.905	4.431	2.449
Min	0	0	2.000	1.083	0.500	0	0.000	0.000	0	0	0	0	0	0	0
Max	192	77	273	242.667	0.905	70	298.833	1.000	1	1	1	1	277	81	27

Notes:
* $p < 0.05$.
** $p < 0.01$.

Hypotheses 2a and 2b predict that an organization's number of direct partners in a collaboration network has inverted U-shaped effect on its exploitative and exploratory innovations. The coefficients of the variable in the reported results of Models 3 and 4 in Table 3 have the expected sign and are significant at $p < 0.01$ for its exploitative innovation. Therefore, Hypothesis 2a confirmed. Moreover, the regression results of Models7 and 8 in Table 3 both support Hypothesis 2b at $p < 0.05$. Thereby, the inverted U-shaped effect of organizations' number of direct partners on their innovation in terms of exploitation and exploration is corroborated by the empirical testing. That is, the initial increases in the number of an organization's direct partners of in a collaboration network tend to favor the organization's exploitative and exploratory innovations. However, when the number of direct partners reaches a certain value, further increases are likely to reduce the organization's exploitative and exploratory innovations. Therefore, the number of direct partners of an organization in a collaboration network influences its exploitative and exploratory innovations in the same way.

Hypotheses 3a and 3b assume that the number of indirect ties of an organization's knowledge elements in a knowledge network facilitate both its exploitative and exploratory innovations. The reported regression results of Models 2 and Model 4 in Table 3 both supports the Hypothesis 3a at $p < 0.01$. By contrast, the regression coefficients of this variable in Models 6 and 8 have the expected positive sign on exploratory innovation, but they do not reach our concerned lowest significant level. Therefore, Hypothesis 3b is rejected. More the indirect ties of an organization's knowledge elements in a knowledge network means more frequent innovations of an organization by repeatedly using its existing knowledge elements by combining or recombining them.

Hypotheses 4a and 4b argue that the number of indirect partners of an organization in a collaboration network hinders its exploitative and exploratory innovations. According to the results reported in Models 3 and 4 in Table 3, the coefficient of this variable is unstable across models. Therefore, this hypothesis is not corroborated. However, the regression results reported in Models7 and 8 in Table 3 both support Hypothesis 4b at $p < 0.01$. As a result, more indirect ties of an organization in a collaboration network results in fewer innovation of the organization by exploring novel knowledge elements. Therefore, the number of indirect ties of an organization in a collaboration network affects its exploitative and exploratory innovations in different ways.

Hypothesis 5a contends that non-redundancy among ties of an organization's knowledge network structures hinders its exploitative innovation. The reported regression coefficients of this variable in Models 2 and 4 in Table 3 have the expected negative sign and are significant at $p < 0.05$ for the organization's exploitative innovation. Therefore, the Hypothesis 5a is accepted. By contrast, Hypothesis 5b contends that non-redundancy among ties of an organization's knowledge network structures facilitates its exploratory innovation. The coefficients of this variable are positive and significant at $p < 0.01$ in Models 6 and 8 in Table 3. Therefore, Hypothesis 5b is also confirmed. An organization's non-redundant knowledge network structure has opposite effects on its exploitative and exploratory innovations.

Hypotheses 6a and 6b predict that non-redundancy among ties of an organization's collaboration network structure facilitates its exploitative and exploratory innovations. The regression results in Models 3 and 4 in Table 3 both support the Hypothesis 6a at $p < 0.01$. Therefore, more non-redundancy of an organization's collaboration network structure results in more frequent innovation of an organization by repeatedly using its existing knowledge elements. In accordance with Hypothesis 6b, the coefficients of this variable in the reported regression results of Models 7 and Model 8 in Table 3 are positive, consistent with our expectation, but they do not reach

Table 3
Random effects panel regression results for exploitative and exploratory innovation.

Variable	Exploitative innovation				Exploratory innovation			
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Knowledge network direct ties								
Mean knowledge direct ties		9.1012*** (.9615)		8.3354*** (.9644)		−.8937 (.5429)		−.8759 (.5478)
(Mean knowledge direct ties)^2		−6.8617*** (.9324)		−6.2194*** (.9274)		−.2573 (.5427)		−.1514 (.5437)
Knowledge network Indirect ties								
Mean distance weighted centrality		3.6316*** (.2737)		3.3046*** (.2944)		.1409 (.4491)		.1306 (.4499)
Knowledge network non-redundancy								
Mean knowledge network efficiency		−2.6084** (1.1666)		−2.6870** (1.1708)		2.2938*** (.7555)		2.2143*** (.7574)
Collaboration network direct ties								
direct ties			4.2622*** (.7005)	4.1000*** (.7321)			2.1983*** (.5823)	1.7640*** (.5838)
(direct ties)^2			−9.0242*** (1.4838)	−6.6903*** (1.5462)			−2.2365** (1.0247)	−2.1128** (1.0423)
Collaboration network indirect ties								
Distance weighted centrality			.5224*** (.0831)	−.1146 (.0914)			−.4856*** (.0714)	−.2598*** (.0770)
Collaboration network non-redundancy								
network efficiency			.2083*** (.0540)	.1421*** (.0548)			.0224 (.0390)	.0185 (.0393)
Control variables								
Asia	.2309*** (.0568)	.2467*** (.0614)	.1613*** (.0567)	.1969*** (.0606)	−.0552 (.0441)	−.0274 (.0436)	−.0490 (.0445)	−.0328 (.0438)
Europe	−.1836** (.0793)	−.1895** (.0834)	−.2417*** (.0789)	−.2408*** (.0819)	.0835 (.0586)	.1090* (.0573)	.0692 (.0587)	.0939 (.0574)
University	−.2423*** (.0545)	−.3251*** (.0581)	−.2276*** (.0538)	−.3224*** (.0567)	.2038*** (.0420)	.2174*** (.0411)	.1955*** (.0420)	.2110*** (.0410)
Institute	.0468 (.0766)	.0258 (.0824)	.0547 (.0753)	.0029 (.0796)	.2848*** (.0595)	.2911*** (.0580)	.2745*** (.0595)	.2843*** (.0579)
Year dummy variables	Included	Included	Included	Included	Included	Included	Included	Included
R&D intensity	.0143*** (.0014)	.0087*** (.0015)	.0095*** (.0015)	.0068*** (.0015)	−.0075*** (.0019)	−.0048*** (.0018)	−.0073*** (.0020)	−.0056*** (.0020)
Exploitative learning ability	.0027 (.0046)	.0069 (.0047)	.0123*** (.0046)	.0114** (.0048)	.0293*** (.0063)	.0273*** (.0061)	.0296*** (.0062)	.0284*** (.0061)
Exploratory learning ability	.0722*** (.0049)	.0926*** (.0051)	.0716*** (.0050)	.0864*** (.0051)	.0502*** (.0058)	.0460*** (.0056)	.0463*** (.0057)	.0441*** (.0056)
Constant	−.7485*** (.0605)	−3.4665*** (1.0049)	−.9848*** (.0675)	−3.1412*** (1.0089)	.1165*** (.0489)	−.6604 (.6425)	.1454*** (.0533)	−.7286 (.6447)
Number of organizations	919	919	919	919	919	919	919	919
Number of observations	5107	5107	5107	5107	5107	5107	5107	5107
Log Likelihood	−10,739.682	−10,520.73	−10,636.238	−10,494.13	−13,517.237	−13,466.918	−13,492.708	−13,459.369

Notes: "Year dummy variables" were included in the models but their coefficients were not reported in the table; standard errors are in the parentheses.
* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.
Standardized variables of direct ties and indirect ties were used in regression estimations.

our concerned lowest significant level. Therefore, Hypothesis 6b is not corroborated.

4.3. Additional analysis

As a robustness check, we next chose to model exploitative and exploratory innovation by zero-inflated negative binomial models. Mode 1 and Model 2 in Table 4 report the zero-inflated negative binomial regression results for exploitative and exploratory innovation respectively. The independent and control variables in these two models are the same as those in Table 3. As illustrated by these two models, our previous findings from the standard negative binomial models are confirmed by zero-inflated negative binomial models.

Except for quality performance in terms of exploitative and exploratory innovations, we also took simple counts for both exploitative and exploratory patents as dependent variables to further test our findings. For quantity performance, we also estimated both standard negative binomial models and zero-inflated negative binomial models. Model 3 through Model 6 in Table 4 report the regression results for quantity performance in terms of exploitative and exploratory innovations. As demonstrated by these models, the signs of the estimated coefficients for our concerned variables are similar to the regression results for quality performance, but the magnitude and the significant level of the estimated coefficients change. In general, findings in Model 3 through Model 6 are consistent with our previous findings.

5. Discussion and conclusions

5.1. Main findings

Technological innovation is embedded not only in social networks but also in knowledge networks (Wang et al., 2014; Yayavaram and Ahuja, 2008). In this study, we examined the structural characteristics of inter-organizational collaboration and knowledge networks in the emerging nano-energy field and their influences on organizations' innovations in terms of exploitation and exploration.

We found that inter-organizational collaboration networks and knowledge networks in the nano-energy field are different in terms of integration. Inter-organizational collaboration networks, which are fragmented across many unconnected clusters, are characterized by a low degree of collaborative integration that provides few opportunities for the diffusion of knowledge and information and joint problem solving. By contrast, knowledge networks in the nano-energy field, in which knowledge elements can reach to each other to a large extent, are characterized by a high degree of associative integration that provides many opportunities for knowledge findings. Moreover, the same structural properties have different meanings and effects in collaboration and knowledge networks (Wang et al., 2014). These two networks are decoupled. An organization's positions in an inter-organizational collaboration network do not necessarily reflect its knowledge elements positions in a knowledge network. This finding is in accordance with that of Wang et al. (2014). For example, an organization with few partners in a collaboration network does not necessarily have knowledge elements with few ties in a knowledge network.

With regard to the role of the number of direct ties, we found that the number of direct ties of an organization's knowledge elements in a knowledge network has a curvilinear effect on its exploitative innovation. An innovative organization clearly benefits from its knowledge elements' direct ties, because direct ties of knowledge elements denote combinatorial potentials. However, up to a certain point, further increases in direct ties in knowledge elements hinders the organization's ability to innovate by reusing its extant knowledge elements, as combinatorial potentials of these knowledge elements are exhausted. Therefore, to facilitate exploitative innovation, the direct ties of an organization's knowledge elements should be maintained at a moderate level. However, we did not corroborate the curvilinear effect of the direct ties of an organization's knowledge elements on its exploratory innovation. The number of direct ties of an organization in a collaboration network has a curvilinear effect on both its exploitative and exploratory innovations. Although increasing the direct partners of an organization in a collaboration network increases its exploitative and exploratory innovations, excessive direct partners impair the organization's competence not only to reuse its existing knowledge elements but also to explore for new ones. Therefore, to facilitate exploitative and exploratory innovations, the ego-network size of an organization in a collaboration network should be maintained at a moderate level.

With regard to the role of the number of indirect ties, we found that the number of indirect ties of knowledge elements of an organization in a knowledge network has a significantly positive effect on its exploitative innovation. Although we failed to find a significant effect of this variable on exploratory innovation, its effect is positive. Therefore, when seeking for innovation opportunities, an organization should conduct not only a local search around its knowledge elements' ego-networks but also a distant search. However, we found that the number of indirect ties of an organization in a collaboration network has a negative effect on its exploratory innovation, but we were unable to find a negative effect of this variable on exploitative innovation.

Finally, with regard to the role of non-redundancy among ties, we found that non-redundancy among ties of knowledge elements of an organization in a knowledge network has opposite effects on its exploitative and exploratory innovations. Therefore, having non-redundant knowledge network structures reduces the organization's ability to innovate by repeatedly using its existing knowledge elements but enhances its ability to innovate by exploring knowledge elements that are novel to its extant stocks. Embedding in open or closed knowledge network structures depends on the specific innovation task of an organization. By contrast, we found that having a non-redundant structure in a collaboration network has a positive effect on exploitative innovation. Although we failed to find a significant effect of this variable on exploratory innovation, a positive effect was observed. Therefore, an organization should be embedded in open collaboration networks to benefit its exploitive and exploratory innovation.

5.2. Contributions

Most of the previous studies considered the knowledge base of an organization to be a simple collection of its knowledge elements. This study introduced knowledge network in the emerging nano-energy field and focused on the influence of knowledge base structure on innovations in terms of exploitation and exploration. A knowledge network consists of a set of knowledge elements or components that do not process agency activities like actors in social networks. The relationship between knowledge elements is associative or combinational, recording the combinative experiences of knowledge elements in previous inventions and representing the results of conscious and deliberate efforts of individuals or higher-level collectives (Wang et al., 2014; Yayavaram and Ahuja, 2008). These associative relationships can serve as a medium of knowledge flow and guidance for future combinations of knowledge elements. For knowledge search, the associative relationships among knowledge elements are even more important than the knowledge elements themselves because combinatorial opportunities or novel knowledge elements can be searched by these ties.

Table 4
Results of zero-inflated negative binomial regression and negative binomial regression.

	Zero-inflated negative binomial regression		Negative binomial regression		Zero-inflated negative binomial regression	
	Exploitative innovation	Exploratory innovation	Number of exploitative patents	Number of exploratory patents	Number of exploitative patents	Number of exploratory patents
Variable	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Knowledge network direct ties						
Mean knowledge direct ties	6.6202*** (.8718)	−.7033 (.6144)	8.8339*** (.9548)	.7484 (.5831)	6.8283*** (.8341)	−.4356 (.5039)
(Mean knowledge direct ties)^2	−5.4362*** (.8236)	−.9002 (.5846)	−6.5636*** (.9183)	−1.1329* (.5162)	−5.1658*** (.7988)	−.5301 (.4922)
Knowledge network Indirect ties						
Mean distance weighted centrality	.7005** (.2759)	.0483 (.1676)	3.7541*** (.2895)	.2208 (.4456)	1.6752*** (.2468)	.1393 (.1339)
Knowledge network non-redundancy						
Mean knowledge network efficiency	−2.4482*** (.3377)	3.6286*** (.2518)	−2.3702** (1.1747)	1.2202* (.7098)	−4.3446*** (.3105)	1.0942*** (.1976)
Collaboration network direct ties						
Direct ties	3.3779*** (.7285)	2.8750*** (.5745)	2.9369*** (.7206)	1.2554** (.6155)	1.2805** (.5500)	.8656** (.4336)
(Direct ties)^2	−5.4108*** (1.0467)	−3.5808*** (.8704)	−4.0420*** (1.5409)	−2.2864* (1.3276)	−2.7521*** (.8152)	−.9644 (.6756)
Collaboration network indirect ties						
Distance weighted centrality	.0350 (.1069)	−.1931** (.0831)	−.1362 (.0874)	−.2707*** (.0739)	−.0611 (.0831)	−.2338*** (.0650)
collaboration network non-redundancy						
Network efficiency	.1639*** (.0550)	.0393 (.0420)	.1909*** (.0540)	.0051 (.0370)	.1036** (.0460)	−.0043 (.0333)
Control variables						
Asia	−.1639*** (.0500)	−.2691*** (.0372)	.1890*** (.0689)	.1225*** (.0384)	.2877*** (.0421)	.1472*** (.0301)
Europe	−.0939 (.0669)	.1766*** (.0500)	−.2999*** (.0936)	−.0485 (.0517)	−.1633*** (.0603)	−.0300 (.0423)
University	−.5289*** (.0480)	−.1575*** (.0360)	−.2949*** (.0639)	.2678*** (.0351)	−.2864*** (.0398)	.2461*** (.0283)
Institute	−.1898*** (.0659)	.0108 (.0502)	−.0137 (.0907)	.2954*** (.0497)	.0115 (.0521)	.2508*** (.0386)
Year dummy variables	Included	Included	Included	Included	Included	Included
R&D intensity	.0303*** (.0032)	.0028 (.0023)	.0052*** (.0014)	−.0016 (.0017)	.0266*** (.0022)	.0046*** (.0016)
Exploitative learning ability	.0478*** (.0109)	.0202** (.0082)	.0128*** (.0041)	.0196*** (.0052)	.0465*** (.0077)	.0117** (.0057)
Exploratory learning ability	.1360*** (.0105)	.0906*** (.0078)	.0768*** (.0047)	.0534*** (.0048)	.1229*** (.0077)	.0910*** (.0054)
Constant	–	–	−2.8509*** (1.0150)	.5479 (.6082)	–	–
Number of organizations	919	919	919	919	919	919
Number of observations	5107	5107	5107	5107	5107	5107
Log Likelihood	−10,766.94	−13,573	−8364.8123	−9353.2764	−8418.975	−9307.81

Notes: "Year dummy variables" were included in the models but their coefficients were not reported in the table; Standard errors are in the parentheses.
* $p < 0.10$.
** $p < 0.05$.
*** $p < 0.01$.
Standardized variables of direct ties and indirect ties were used in regression estimations.

Most of the previous studies investigated how social relationships and the network they formed influence innovation. The present study highlights multiple network embeddednesses of organizations in exploitative and exploratory innovation activities. A technology-based collaboration network is formed by a set of agents such as individuals or higher-level collectives, which can search for and transmit knowledge and information and also create novel ones (Phelps et al., 2012). These agents are interconnected by collaborative relationships that enable and constrain the diffusion, transfer and acquisition of knowledge and information and the subsequently innovative abilities (Phelps et al., 2012; Vanhaverbeke et al., 2006). The collaborative relationship between these agents is social, including formal or informal collaboration, which is conscious and deliberate agency activities (Ahuja et al., 2012). These social relationships can serve as a means through which agents can search for knowledge and information, a channel by which knowledge and information flow, and a lens by which social agents evaluate other actors and their knowledge and information stocks (Phelps et al., 2012). As discussed above, knowledge network mirrors how knowledge elements are connected to or separated from one another. These two types of networks are decoupled, with the same network property having distinct meanings and influences.

Most of the previous studies especially emphasized the role of social-based search on innovation. This study goes beyond the dominant focus and highlights the role of knowledge-based and social-based searches. Knowledge-based search reinforces and complements social-based search. Social-based search emphasizes the importance of social linkages through which knowledge and information owned by other agents can be searched. Knowledge-based search highlights the importance of knowledge linkages through which combinatorial opportunities or novel knowledge elements can be searched.

5.3. *Limitations and future research*

Although this study has significant theoretical and practical implications, it also has several limitations that could be explored in future research. Exploitative and exploratory innovations have been calculated in different approaches in the extant literature. In this paper, we defined an organization's exploitative and exploratory innovations based on patents whether categorizing by new technological classes by comparing the organization's extant technology classes. Some other studies defined an organization's exploitative and exploratory innovations based on patents whether citing its other prior patents. These two approaches measure exploitative and exploratory innovations in different ways. An organization may patent in new technology classes to its existing stock but with references to its prior patents; the opposite is also possible. Therefore, future research may combine these two approaches to obtain more precise measures for exploitative and exploratory innovations. Moreover, to overcome the limitation of simple patent count, we considered the family size as weight to measure innovation. Other studies used the number of forward citations as weight. Therefore, the forward citation weighted innovation measures may be used in future research.

Our empirical analysis is limited to only one industrial context, that is, the emerging nano-energy field, which is characterized by high dynamism and interactions across disciplines. Therefore, our results cannot be simply generalized to other industrial contexts. However, the research design in this paper should be replicated. Other industrial contexts, which are more or less dynamic and interacting, should be further studied, especially to test the hypotheses that we cannot confirm in the present research. Furthermore, the unit of analysis in this paper is at the organizational level, which can be shifted to the level of individual researchers in future studies.

An organization's knowledge base is an important factor that should be considered when establishing technological collaboration relationships. Future research may investigate the effect of combinational relationships between knowledge elements and their corresponding networks on technological collaborations.

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Appendix 1.

See Table A1.

Table A1
Definition of search queries for patents of nano-energy technology in DII database.

Searching sets	Searching terms
Nano-technology	**#1** TS = (nano*)
	#2 TS = ((“quantum dot*” OR “quantum well*” OR “quantum wire*”) NOT nano*)
	#3 TS = (((“self assembl*” OR “self organiz*” OR “directed assembl*”) AND MolEnv-I) NOT nano*)
	#4 TS = ((“molecul* motor*” OR “molecul* ruler*” OR “molecul* wir*” OR “molecul* devic*” OR “molecular engineering” OR “molecular electronic*” OR “single molecul*” OR fullerene* OR buckyball OR buckminsterfullerene OR C60 OR “C-60” OR methanofullerene OR metallofullerene OR SWCNT OR MWCNT OR “coulomb blockad*” OR bionano* OR “Langmuir–Blodgett” OR Coulombstaircase* OR “PDMS stamp*” OR graphene OR “dye-sensitized solar cell” OR DSSC OR ferrofluid* OR “core-shell”) NOT nano*)
	#5 TS = ((((TEM or STM or EDX or AFM or HRTEM or SEM or EELS or SERS or MFM) OR “atom* force microscop*” OR “tunnel* microscop*” OR “scanning probe microscop*” OR “transmission electron microscop*” OR “scanning electron microscop*” OR “energy dispersive X-ray” OR “xray photoelectron*” OR “x-ray photoelectron” OR “electron energy loss spectroscop*” OR “enhanced raman-scattering” OR “surface enhanced raman scattering” OR “single molecule microscopy” OR “focused ion beam” OR “ellipsometry” OR “magnetic force microscopy”) AND MolEnv-R) NOT nano*)
	#6 TS = (((NEMS OR Quasicrystal* OR “quasi-crystal*” OR “quantum size effect” OR “quantum device”) AND MoleEnv-I) NOT nano*)
	#7 TS = (((biosensor* OR NEMS OR (“sol gel*” OR solgel*) OR dendrimer* OR CNT OR “soft lithograph*” OR “electron beam lithography” OR “e-beam lithography” OR “molecular simul*” OR “molecular machin*” OR “molecular imprinting” OR “quantum effect*” OR “surface energy” OR “molecular sieve*” OR “mesoporous material*” OR “mesoporous silica” OR “porous silicon” OR “zeta potential” OR “epitax*”) AND MolEnv-R) NOT nano*)
	#8 TS = (plankton* OR n*plankton OR m*plankton OR b*plankton OR p*plankton OR z*plankton OR nanoflagel* OR nanoalga* OR nanoprotist* OR nanofauna* OR nano*aryote* OR nanoheterotroph* OR nanophtalm* OR nanomeli* OR nanophyto* OR nanobacteri* OR *270 organism names beginning with nano* OR nano2 OR nano3 OR nanos OR nanog OR nanor OR nanoa OR nano- OR nanog- OR nanoa- OR nanor- OR nanosatellite* OR 270 organism names beginning with nano*)
	# 9 TS = (nanometer* OR nano-meter OR nano-meter OR nano-meter OR nanosecond* OR nano-second OR nanomolar* OR nano-molar OR nanomole(s) OR nanogram* OR nano-gram OR nanoliter* OR nanolitre* OR nano-liter OR nano-liter*)
	Total Nanotechnology = ((#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7) NOT #8 NOT (#9 NOT (#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7)))

Table A1 (*Continued*)

Searching sets	Searching terms
Energy	**# 1** TS = (energ* SAME (“energy sector” OR “power source*” OR renewable* OR “power supply” OR “energy convers*” OR “energy storag*” OR sustainab* OR use* OR “distribution loss*” OR harvest* OR “wind energy” OR eolic OR tidal OR biomas* OR geotherm* OR hydroelectric* OR (wave SAME (ocean OR sea)) OR fos$il OR oil OR “natural gas” OR coal OR “nuclear energy” OR fuel OR tide OR petroleum OR heat$storag* OR thermal$insulator OR batter* OR super$capacitor* OR capacitor* OR flywheel* OR photovoltaic* OR “solar cell*” OR power*station OR carge$carrier OR “fuel cell*” OR electro$catalys* OR photoelectrochem* OR thermo$electric* OR turbin* OR transducer* OR “water photoelectrolys*” OR power$generat* OR biofuel* OR biodiesel* OR water$oxidation OR “combustion engine” OR “thermal rectifier” OR “hydrogen* production”)) **#2** TS = (photosynthesis OR obesity OR diet* OR food OR cellular OR glucose OR DNA OR astrophys* OR astronom* OR chlorop* OR phyto*) OR SO = (astrophys* OR astronom* OR biolog* OR nutriti* OR botanic* OR American Journal of Clinical Nutrition OR Monthly Notices of the Royal Astronomical Society OR Biotechnology and Bioengineering OR Annual Review of Nutrition OR Journal of Geophysical Research-Space Physics) **Total Energy** = # 1 NOT #2
Nano-enery	**Total Nano-energy** = Total Nanotechnology AND Total Energy

Note: Searching terms for nanotechnology came from Arora et al. (2013). Given spatial limitations, the details of “MolEnv-I”, “MolEnv-R” and “270 organism names beginning with nano*” were not listed in this table, but they all can be found in the article of Arora et al. (2013); And searching terms for energy mainly came from Menéndez-Manjón et al. (2011). We carried on some modification and supplement for energy retrieval terms based on energy searching words used by Connelly and Sekhar (2012), Lee et al. (2012) and Lee and Lee (2013).

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